



Lambda vs EC2



What is AWS Lambda?

- Lambda is a serverless compute service where you run code without provisioning or managing servers. It's event-driven – great for API calls, file uploads, scheduled tasks, and more. You only pay when your code runs, and it scales automatically.

What is Amazon EC2?

- EC2 (Elastic Compute Cloud) gives you a virtual machine in the cloud. You control everything – OS, runtime, packages. It's ideal for apps that need longer runtime, persistent connections, or custom environments. But with control comes responsibility – you handle scaling, security patches, etc.

Lambda vs EC2 - Differences

Feature	Lambda	EC2
Server management	Fully managed	You manage it
Pricing model	Pay per request time used	Pay for uptime
Scalability	Auto-scaled	Manual or Auto-scaled
Runtime limit	15 minutes	No limit
Customization	Limited	Full control
Best for	APIs, Event triggers	Web servers, databases, long jobs

When to Use Lambda?

- Event-driven workflows (S3 upload triggers, API Gateway)
- Lightweight microservices
- Scheduled tasks (cron jobs)
- Apps that need auto-scaling and low cost

When to Use EC2?

- Long-running applications
- Complex backend systems needing more control
- Apps with specific OS/runtime requirements
- Self-hosted databases, containers, or frameworks



AWS Lambda – Pros and Cons

Pros

- Quick to deploy
- Auto-connects with other AWS services
- No server management
- Good for small, focused tasks

Cons

- Debugging can be tricky
- Slower start if inactive (cold start)
- Limited control over environment
- Harder to test locally

Amazon EC2 – Pros and Cons

Pros

- Full control over setup
- Runs long and complex apps
- Good for traditional software
- Flexible hardware choices

Cons

- Manual setup and updates
- Slower to launch new projects
- Costs run even when idle
- Security setup needs care



Thank
You!

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